

# Logistics Performance Report: Nassau Candy Distribution

This report analyses shipping performance and route efficiency for Nassau Candy across major manufacturing hubs and regional destinations. The data evaluates lead times, shipment volumes, and consistency to identify operational strengths and bottlenecks.

---

## 1. Executive Summary

The distribution network is powered by five primary manufacturing facilities: **Lot's O' Nuts**, **Secret Factory**, **Sugar Shack**, **The Other Factory**, and **Wicked Choccy's**. Performance is characterized by significant regional variance, with lead times ranging from highly efficient short-duration shipments to notable outliers.

### Key Network Metrics:

- **Primary Distribution Hub:** California serves as the highest-volume destination, receiving over 1,900 shipments from Lot's O' Nuts and Wicked Choccy's alone.
  - **Operational Benchmark:** The **Secret Factory** maintains the highest efficiency levels across the network, particularly on routes to the Midwest and South.
  - **Consistency Challenge:** Routes to **British Columbia** exhibit the highest standard deviation (307.08), indicating a need for more predictable transit processes.
- 

## 2. Factory Performance Deep-Dive

### A. High-Volume Facilities (Lot's O' Nuts & Wicked Choccy's)

- **Scale:** Lot's O' Nuts is the network's workhorse, managing 1,125 shipments to California with an average lead time of 1,316.9 units.
- **Efficiency:** Wicked Choccy's maintains similar performance levels, with heavy volume in California (823 shipments) and New York.

### B. Specialized Facilities (Secret Factory & The Other Factory)

- **Peak Efficiency:** Secret Factory is a leader in speed, achieving near-perfect efficiency scores (0.99+) on routes to **Delaware, Minnesota, Mississippi, Nebraska, and New Mexico**.

- **Optimization Targets:** The Other Factory shipments to **North Carolina** and **Nevada** show critical inefficiency (scores below 0.01), marking them as high-priority areas for logistics restructuring.

---

### 3. Regional KPI Analysis

The following table summarizes the highest and lowest-performing routes based on the **Efficiency Score** (where 1.0 is the network's best performance):

| Route Origin      | Destination  | Total Shipments | Avg. Lead Time | Efficiency Score |
|-------------------|--------------|-----------------|----------------|------------------|
| Secret Factory    | Nebraska     | 1               | 906.0          | 1.000            |
| Secret Factory    | New Mexico   | 2               | 906.0          | 1.000            |
| The Other Factory | Louisiana    | 1               | 907.0          | 0.998            |
| Wicked Choccy's   | North Dakota | 2               | 1637.0         | 0.006            |
| Lot's O' Nuts     | North Dakota | 5               | 1638.2         | 0.005            |
| Sugar Shack       | New Jersey   | 1               | 1642.0         | 0.000            |

---

### 4. Methodology & Data Integrity

- **Lead Time Calculation:** Defined as the calendar days between the Order Date and the Ship Date.
  - **Efficiency Scoring:** A normalized index where 1.0 represents the minimum average lead time in the dataset and 0.0 represents the maximum.
  - **Delay Tracking:** Any shipment with a lead time exceeding 5 days is flagged as "Delayed" for internal auditing.
-

## 5. Strategic Recommendations

1. **Midwest Transit Audit:** Investigate why North Dakota routes consistently rank as the least efficient across multiple factories.
2. **Secret Factory Model:** Adopt the logistics and carrier selection strategies used by the Secret Factory for other low-volume, high-efficiency routes.
3. **Carrier Performance Review:** Address the 0.0 efficiency score for the *Sugar Shack* → *New Jersey* lane, which represents the longest lead time in the current network.